Ultra sonar code with alert:

```
int trigPin = 9;    // TRIG pin
int echoPin = 8;    // ECHO pin

float duration_us, distance_cm;

void setup() {
// put your setup code here, to run once:
Serial.begin(9600);
Serial.println("hello world");
// configure the trigger pin to output mode
pinMode(trigPin, OUTPUT);
// configure the echo pin to input mode
pinMode(echoPin, INPUT);
}

void loop() {
// put your main code here, to run repeatedly:
digitalWrite(trigPin, HIGH);
delay(1000);
digitalWrite(trigPin, LOW);

// measure duration of pulse from ECHO pin
duration_us = pulseIn(echoPin, HIGH);

// calculate the distance
distance_cm = 0.017 * duration_us;

// print the value to Serial Monitor
Serial.print("distance: ");
Serial.print(distance_cm);
Serial.println(" cm");
if (distance_cm<7) {
Serial.print("alert flood might occur ");
// do stuff if the condition is true
}
```

```
delay(500);


}
```

New try:

```
#include <SoftwareSerial.h>

// Define pins for SoftwareSerial
SoftwareSerial gsmSerial(2, 3); // RX, TX
int trigPin = 9;    // TRIG pin
int echoPin = 8;    // ECHO pin

float duration_us;
float distance_cm=100;
int count=0;

void setup() {
// Start communication with Arduino Serial Monitor
Serial.begin(9600);
Serial.println("welcome to water level monitoring system");
// configure the trigger pin to output mode
pinMode(trigPin, OUTPUT);
// configure the echo pin to input mode
pinMode(echoPin, INPUT);



}

void loop() {
digitalWrite(trigPin, HIGH);
delay(1000);
digitalWrite(trigPin, LOW);

// measure duration of pulse from ECHO pin
duration_us = pulseIn(echoPin, HIGH);
```

```
// calculate the distance
distance_cm = 0.017 * duration_us;

Serial.print("distance: ");
Serial.print(distance_cm);
Serial.println(" cm");

if (distance_cm<10 && count==5) {
Serial.print("alert flood might occur ");
count=count+1;
Serial.print("count=");
Serial.print(count);
Serial.println("Initializing GSM module...");

// Start communication with GSM module
gsmSerial.begin(9600);
delay(1000);

// Test GSM module
gsmSerial.println("AT"); // Check communication
delay(1000);
if (gsmSerial.available()) {
Serial.println("GSM Module Response:");
while (gsmSerial.available()) {
Serial.write(gsmSerial.read());
}
}

// Set SMS Text Mode
gsmSerial.println("AT+CMGF=1"); // Set GSM module to text mode
delay(1000);

// Specify phone number and send the message
gsmSerial.println("AT+CMGS=\"+8801712167018\""); // Replace with recipient's phone number
delay(1000);
gsmSerial.print("flood started"); // Message content
```

```
delay(1000);
gsmSerial.write(26); // ASCII code for CTRL+Z (end of SMS command)
delay(1000);

Serial.println("Message sent!");}
// do stuff if the condition is true

// Keep checking for GSM module responses
if (gsmSerial.available()) {
Serial.write(gsmSerial.read());
}
}
```

Ultra + GSM with count:

```
#include <SoftwareSerial.h>

// Define pins for SoftwareSerial
SoftwareSerial gsmSerial(2, 3); // RX, TX
int trigPin = 9;            // TRIG pin
int echoPin = 8;            // ECHO pin

float duration_us;
float distance_cm = 100;
int count = 0;

void setup() {
 // Start communication with Arduino Serial Monitor
 Serial.begin(9600);
 Serial.println("Welcome to Water Level Monitoring System");

 // Configure the trigger and echo pins
 pinMode(trigPin, OUTPUT);
 pinMode(echoPin, INPUT);

 // Start GSM module
 gsmSerial.begin(9600);
 delay(1000);
```

```
  gsmSerial.println("AT");
  delay(1000);

  // Check if GSM module is responsive
  if (gsmSerial.available()) {
   while (gsmSerial.available()) {
    Serial.write(gsmSerial.read());
   }
   Serial.println("GSM Module Initialized!");
  } else {
   Serial.println("GSM Module not responding!");
  }

  // Set GSM module to SMS Text Mode
  gsmSerial.println("AT+CMGF=1"); // Text mode for SMS
  delay(1000);
}

void loop() {
  // Trigger the ultrasonic sensor
  digitalWrite(trigPin, LOW);
  delay(100);
  digitalWrite(trigPin, HIGH);
  delay(100);
  digitalWrite(trigPin, LOW);

  // Measure the duration of the ECHO pulse
  duration_us = pulseIn(echoPin, HIGH);

  // Calculate the distance in cm
  distance_cm = 0.017 * duration_us;

  Serial.print("Distance: ");
  Serial.print(distance_cm);
  Serial.println(" cm");
```

```
  // Check if distance is below threshold
  if (distance_cm < 10)
  { count++;
    if(count >8) {

    Serial.println("Alert! Flood might occur.");
    Serial.print("Count = ");
    Serial.println(count);
    count=0;
    // Send SMS
    gsmSerial.println("AT+CMGS=\"+8801934901212\""); // Replace with recipient's number
    delay(1000); // Wait for '>'
    gsmSerial.print("Flood warning: Water level critical!");
    gsmSerial.write(26); // ASCII code for CTRL+Z
    delay(5000); // Wait for SMS transmission

    Serial.println("Message sent!");

  }



  }
  else {count=0;}

  // Handle GSM responses
  while (gsmSerial.available()) {
    Serial.write(gsmSerial.read());
  }

  delay(2000); // Avoid frequent checks
}
```

LCD

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27,20,4);  // set the LCD address to 0x27 for a 20x4 LCD display

void setup()
{
  lcd.init();                  // Initialize the LCD
  lcd.backlight();             // Turn on the backlight

  // Print messages to the LCD
  lcd.setCursor(3,0);
  lcd.print("Hello, R1F!");
  delay(3000);
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("  Welcome to");
  delay(2000);
  lcd.setCursor(11,1);
  lcd.print("water level monitoring system");
}

void loop()
{
  // Scroll text left
  for (int i = 0; i < 67; i++) { // Adjust '20' to match the number of scroll steps
    lcd.scrollDisplayLeft();    // Scroll the display to the left
    delay(350);               // Adjust delay for scrolling speed
  }

  // Optional: Scroll text back to the original position
  //for (int i = 0; i < 20; i++) {
    //lcd.scrollDisplayRight();    // Scroll the display to the right
    //delay(250);               // Adjust delay for scrolling speed
```

```
 // }
}
```

UNO+LCD+ULTRA:

```
#include <SoftwareSerial.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// Pins for SoftwareSerial and Ultrasonic Sensor
SoftwareSerial gsmSerial(2, 3); // RX, TX
int trigPin = 9;            // TRIG pin
int echoPin = 8;            // ECHO pin

// LCD Configuration (20x4 display)
LiquidCrystal_I2C lcd(0x27, 20, 4);

// Variables
float duration_us;
float distance_cm;
int count = 0;
const float THRESHOLD = 10.0; // Critical distance in cm

void setup() {
 // Initialize Serial Monitor
 Serial.begin(9600);
 Serial.println("Welcome to Water Level Monitoring System");

 // Initialize ultrasonic sensor pins
 pinMode(trigPin, OUTPUT);
 pinMode(echoPin, INPUT);

 // Initialize GSM module
 gsmSerial.begin(9600);
 delay(1000);
 initializeGSM();

 // Initialize LCD
```

```
  lcd.init();
  lcd.backlight();
  lcd.setCursor(0,0);
  lcd.print("Hello, R1F! kutta");
  delay(3000);
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("Welcome to water");
  lcd.setCursor(0,1);
  lcd.print("level monitoring system");
  delay(4000);
  lcd.clear();
}

void loop() {
  // Measure distance
  distance_cm = getDistance();
  Serial.print("Distance: ");
  Serial.print(distance_cm);
  Serial.println(" cm");
  // Display the distance on the LCD
  lcd.setCursor(0, 0);
  lcd.print("Distance:");
  lcd.print(distance_cm);
  lcd.print("cm"); // Clear any leftover characters

  // Check if water level is critical
  if (distance_cm < THRESHOLD) {
   count++;
   if (count > 8) {
    Serial.println("Alert! Flood might occur.");
    lcd.setCursor(0, 1);
    lcd.print("Flood Warning!    "); // Clear leftover characters
    sendAlert();
    count = 0; // Reset the counter after sending alert
   }
```

```
  } else {
    count = 0; // Reset the counter if water level is safe
    lcd.setCursor(0, 1);
    lcd.print("Safe Water Level   "); // Clear leftover characters
  }



  // Handle GSM responses
  while (gsmSerial.available()) {
    Serial.write(gsmSerial.read());
  }

  delay(1000); // Update every second
}

// Function to measure distance
float getDistance() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  duration_us = pulseIn(echoPin, HIGH);
  return 0.017 * duration_us; // Convert to cm
}

// Function to initialize GSM module
void initializeGSM() {
  for (int i = 0; i < 5; i++) {
    gsmSerial.println("AT");
    delay(1000);
    if (gsmSerial.available()) {
      while (gsmSerial.available()) {
        Serial.write(gsmSerial.read());
      }
```

```
    Serial.println("GSM Module Initialized!");
    return;
   }
   Serial.println("Retrying GSM Initialization...");
  }
  Serial.println("GSM Module not responding. Check connections.");
}

// Function to send an SMS alert
void sendAlert() {
  Serial.println("Alert! Flood might occur.");
  gsmSerial.println("AT+CMGS=\"+8801934901212\""); // Replace with recipient's number
  delay(1000); // Wait for '>'
  gsmSerial.print("Flood warning: Water level critical!");
  gsmSerial.write(26); // CTRL+Z to send the message
  delay(5000);
  Serial.println("Message sent!");
}
```